

STATEMENT OF WORK

For

Automated Smoke Point Instrument

05 March 2026

1. Background:

The Aerospace Fuels Laboratory at Vandenberg Space Force Base (AFPET/PTPLE) provides laboratory quality surveillance and investigative analytical support of flight critical products directly related to the warfighter and space launch operations on behalf of all Air Force and other DoD agencies, as well as commercial entities. AFPET/PTPLE has a vision to create a flexible, dynamic center of excellence that is responsive to customers and strategically positioned and focused to contingency and warfighter operations. AFPET/PTPLE provides the warfighter and space launch activities specialized testing services and capabilities in petroleum, propellants, cryogenics, chemicals and gases for all aerospace vehicles, systems, and equipment.

2. Scope:

AFPET/PTPLE has a requirement for one (1) Automated Smoke Point Instrument to perform smoke point testing on Aviation Turbine Fuels and Kerosene-based Propellants. The instrument shall be a robust unit for routine specification and/or investigative testing. The unit shall comply with the requirements set forth in ASTM D1322-24, *Smoke Point of Kerosene and Aviation Turbine Fuel*.

Copies of ASTM Methods may be obtained by writing to ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959; or by telephone at 610-832-9255; or by e-mail at service@astm.org; or through the ASTM website at www.astm.org

3. Minimum Technical Requirements:

The Automated Smoke Point Instrument shall, at a minimum, meet **all** of the following requirements:

- 3.1 Shall be one (1) instrument (with any strictly necessary add-ons) capable of automatically running ASTM D1322 and shall meet the following requirements:
 - 3.1.1. Configured for standard 100-240V single phase, 50/60Hz electricity.
 - 3.1.1.1. Equipped with 3 wire electrical cord with 20 amp plug.
 - 3.1.2. Benchtop unit, Weight ≤ 50 lbs., WxDxH $\leq 24'' \times 24'' \times 24''$ to fit within designated location without reconfiguration of lab.
 - 3.1.3. Ability to be used in analysis of Aviation Turbine Fuel and other Kerosene-based fuels.

- 3.1.4. Equipped with digital camera connected to a computer to analyze and record the height of the flame.
 - 3.1.5. Ability to read flame heights ranging from 0.0 to 50.0 mm.
 - 3.1.5.1. Shall have a minimum resolution of 0.05 mm for the flame height.
 - 3.1.6. Equipped with temperature, pressure & humidity sensors. Sensors shall meet or exceed the following range & accuracy requirements:
 - 3.1.6.1. Pressure: 30 to 110 kPa with accuracy of ± 0.5 kPa.
 - 3.1.6.2. Humidity: 10 to 90% RH.
 - 3.1.6.3. Temperature: 5 to 50 °C.
 - 3.1.7. Equipped with a candle displacement system to adjust the height of the flame.
 - 3.1.8. Ability to automatically compensate for variable atmospheric conditions during analysis.
 - 3.1.9. Ability to automatically calculate the correction factor, f , according to Equation 1 in ASTM D1322.
 - 3.1.10. Equipped with digital display screen to show operating methods, conditions/settings and results.
 - 3.1.11. Ability to store data & results with >2GB storage;
 - 3.1.11.1. Equipped with USB 2.0+, Ethernet & RS-232 ports.
 - 3.1.12. Ability to connect unit to printer to print results.
- 3.2. Instrument package shall come with:
 - 3.2.1. One (1) Wick Trimmer Set.
 - 3.2.2. One (1) Wick Tube.
 - 3.2.3. 50 compatible woven circular cotton extracted Wicks.
 - 3.2.4. Wick specifications:
 - 3.2.4.1. Casing: 17 ends, 66 tex by 3.
 - 3.2.4.2. Filling: 9 ends, 100 tex by 4.
 - 3.2.4.3. Weft: 40 tex by 2.
 - 3.2.4.4. Picks: 6 per centimeter.
 - 3.3. On-site installation, performance verification, and training of instrument at Vandenberg SFB, CA.
 - 3.3.1. A minimum of four (4) hours of training, not including installation time, to be given.
 - 3.4. Provide technical support for instrument.
 - 3.4.1. Assistance in identifying, troubleshooting and resolving hardware and software problems.
 - 3.4.2. Ability to contact by phone or e-mail during business hours, M-F 9-5pm (nominal).
 - 3.4.2.1. Response time of no more than 3 business days for technical support requests.
 - 3.5. Shall include electronic copies of all manuals and instructions for instrument operation.

4. Delivery, Installation, and Lead Time:

The Contractor shall deliver and install the product within 12 weeks after contract award to the following address:

SHIP TO DoDAAC: FP2075
Aerospace Fuels Laboratory (FP2075)
AFPET/ PTPLE
1747 Utah Ave Bldg 6670
Vandenberg AFB CA 93437-5220

5. Warranty:

The Contractor shall include at least a two (2) year extended warranty for original equipment manufacturer parts and services.

6. Invoice and Payment:

The contractor shall be paid upon submission of a properly prepared invoice and after final inspection and acceptance has been made. A proper invoice shall cite the contractor's name and address, invoice number, invoice date, order number, contract line-item number (CLIN), description of the supplies or service, unit of issue, unit price and extended price, terms of any prompt payment discount offered, and who to notify in the event the invoice is defective. The invoice shall be submitted through Wide Area Work Flow (WAWF) in accordance with the terms of the final contract award.

7. Delivery Procedures – Commercial Vehicles:

- a. The deliverer shall properly ship the system to the Aerospace Fuels Laboratory at Vandenberg AFB CA. All shipping costs to include insurance will be included in contractor's proposal. Great care using standard, approved packaging methods shall be taken.
- b. All vehicles larger than a large pick-up truck are required to be inspected by the Vandenberg Space Force Base Commercial Vehicle Delivery Gate (CVDG) prior to entering the installation. Vehicles to be inspected include, but are not limited to, the following:
 1. Step van/panel truck
 2. Tractor/trailer, box and flatbed containing cargo
 3. Tanker trucks
 4. Box trucks
 5. Tour buses
 6. Garbage/recycled waste trucks
 7. Concrete trucks/mixers, dump trucks
 8. Cranes, recreational vehicles, petroleum tanker
- c. This inspection will be conducted at the CVDG gate (Lompoc Gate).

1. If the vehicle has the product inside (concrete and asphalt trucks) and timely delivery is necessary due to product deterioration it does not need to enter the CVDG. To bypass the CVDG, the contractor shall submit a list containing drivers' names, social security numbers and the state in which the driver's license is held for those drivers who will be entering the base. This shall be accomplished 24 hours prior to requested entry time. If the driver's name is not on the list, he/she will not be allowed access to the installation through these gates and the base will not assume liability for denied access.
2. All irregular deliveries will be sent for inspection. All visitors, which include contractors and commercial vendors without proper background checks and contractor passes, will not be allowed.